



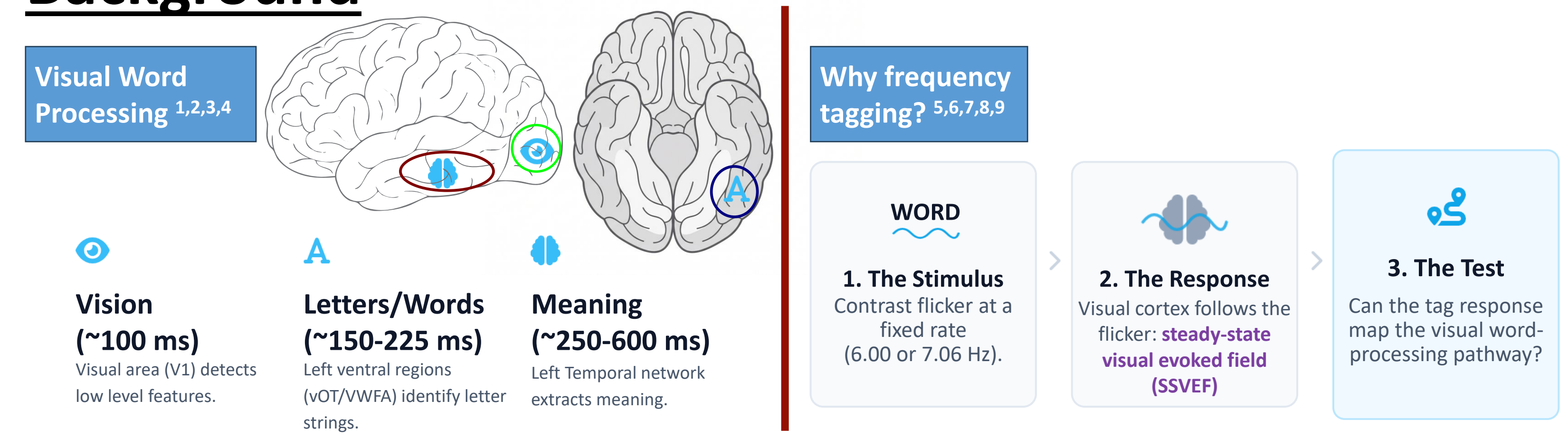
Characterizing word processing pathway using frequency tagging

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A **6 / 7.06 Hz flicker tag** tracks the visual input, not wordness; the **word-locked response** (250–600 ms) separates words from consonant strings, and in anterior temporal cortex the difference is sustained across words.

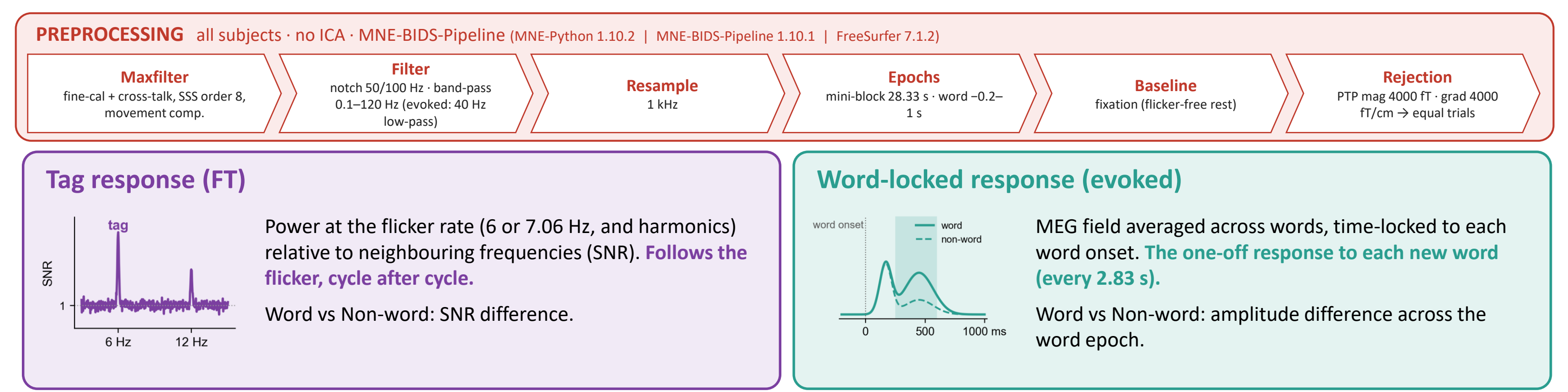
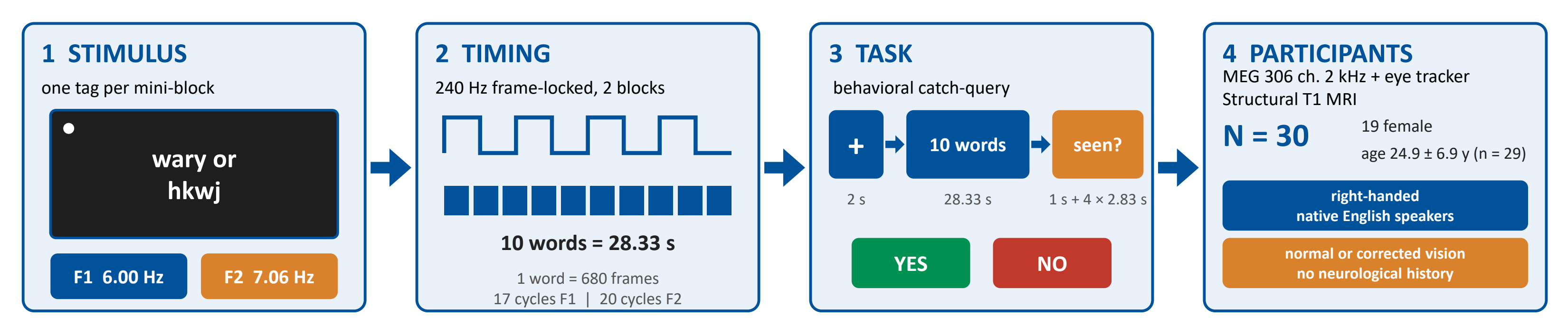
Update vs. submitted abstract: with fixation baselines and equal trial counts, the tag shows no word-specific effect and the source effect is Word > Non-word.

Background



- AIM 1**
Do MEG sensors and sources recover the V1 → vOT → temporal sequence of the literature?
- AIM 2**
Does the tag response (SSVEF) differ between words and consonant strings?

Methods



Pitfalls we hit — and how we fixed them

- Tags off by one FFT bin.** The display ran ~0.5 % slow (photodiode: 5.965 / 7.024 Hz): read the SNR at the measured bin.
- Unequal trial counts faked an effect.** A time-resolved source effect vanished once counts were equalized (the whole-epoch effect remains).
- No gap between words.** The pre-stimulus window holds the previous word: baseline on fixation instead.
- Check the noise-covariance rank.** After SSS, 3 subjects kept a ~1e-38 eigenvalue (dSPM × 10⁶, distorted maps): fixed with mne.compute_rank(tol='auto').



Code · supplementary figures S1–S4 · cohort
github.com/Orion212-lab/snl2026-ft-word-pathway

Cohort 30 recorded; each panel states its N. Excluded: no flicker on the photodiode, malformed blocks, all epochs rejected; sources: no T1 MRI, artifact (sub-20/24). Kept with trigger anomalies: sub-10–14 (main results unchanged without them). List via QR.

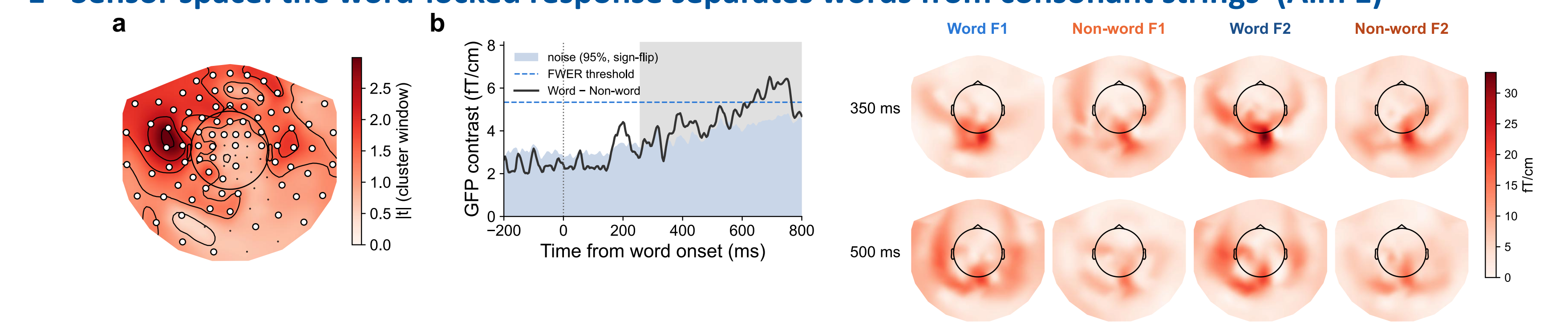
A priori Word vs Non-word at the tag and in the evoked cascade windows (80–130, 150–225, 250–600 ms), left-temporal sensors. Not preregistered; everything else is exploratory.

Limits consonant strings differ in orthography, pronounceability, visual complexity, difficulty · condition-pure blocks · late vOT fragile · localizer, catch-query accuracy not yet analysed

Next pseudowords · item-rate (FPVS) tagging instead of contrast flicker · interleaved conditions · two-word phrase task (intermodulation), recorded, not yet analysed

Results

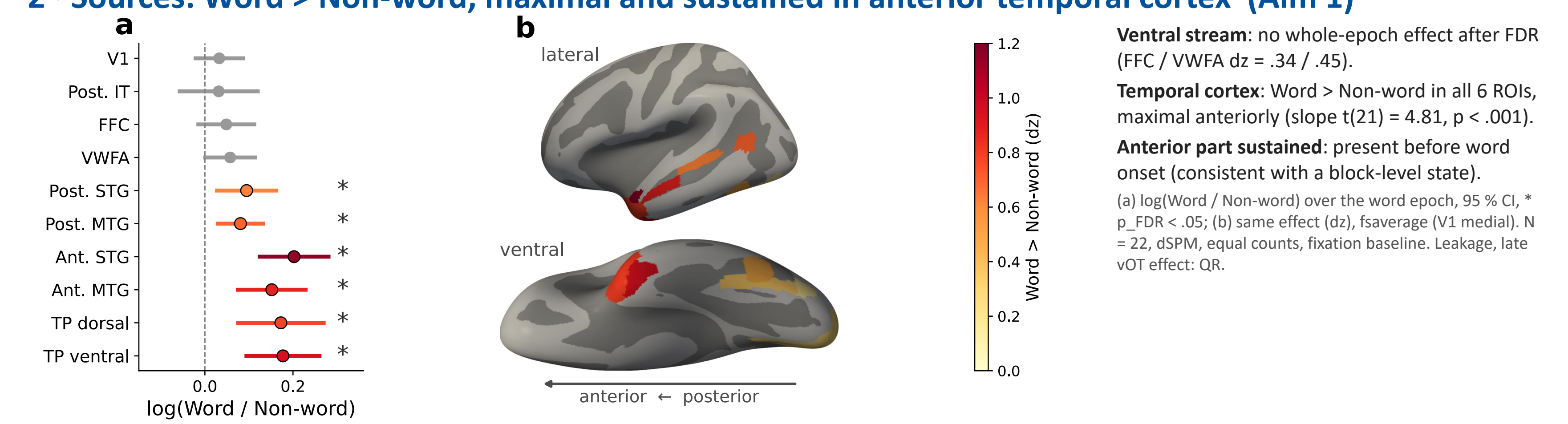
1 · Sensor space: the word-locked response separates words from consonant strings (Aim 1)



One Word > Non-word cluster over left posterior and temporal sensors (N = 26, 84 sensors, $p < .001$), robust to blinks, saccades and artifacts (clean epochs: $dz = 1.31$, 20/21 subjects). Tag × Wordness: n.s.

(a) mean $|t|$ in the cluster window (N = 26); (b) GFP of W - NW vs sign-flip noise (shaded: 95 %; dashed: FWER), exceeded at 628–764 ms; right: grand averages (N = 25) at 350 / 500 ms, ±25 ms; fixation baseline.

2 · Sources: Word > Non-word, maximal and sustained in anterior temporal cortex (Aim 1)



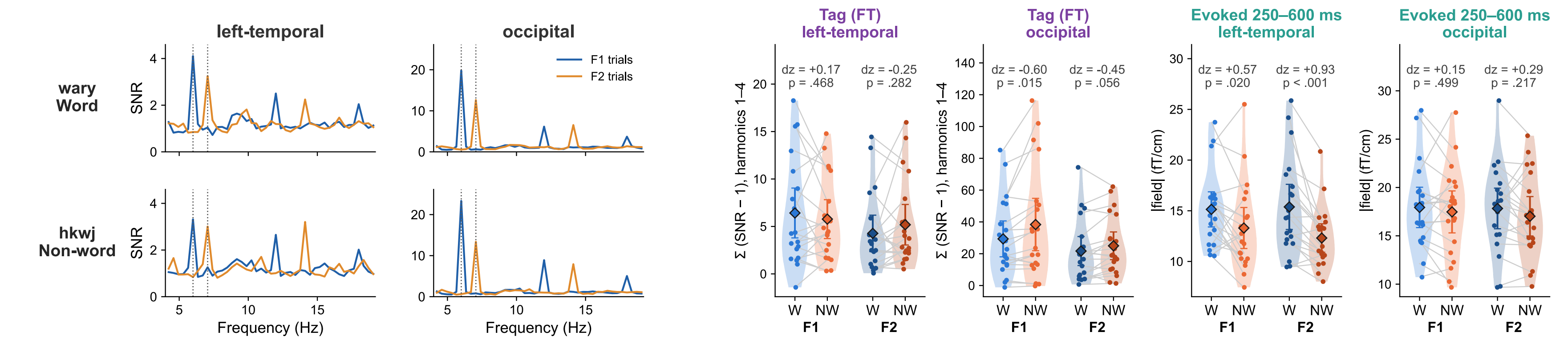
Ventral stream: no whole-epoch effect after FDR (FFC / VWFA $dz = .34 / .45$).

Temporal cortex: Word > Non-word in all 6 ROIs, maximal anteriorly (slope $t(21) = 4.81$, $p < .001$).

Anterior part sustained: present before word onset (consistent with a block-level state).

(a) $\log(\text{Word} / \text{Non-word})$ over the word epoch, 95 % CI, * $p_{\text{FDR}} < .05$; (b) same effect (dz), fsaverage (V1 medial). N = 22, dSPM, equal counts, fixation baseline. Leakage, late vOT effect: QR.

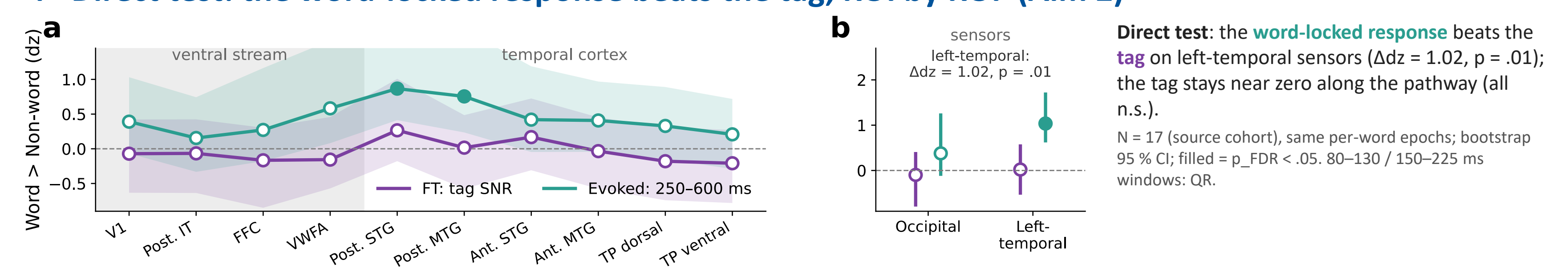
3 · The tag follows the visual input, not wordness (Aim 2)



Tag (harmonics 1–4): occipital Non-words > Words ($dz = -.88$, $p < .001$), following their larger letter area on screen; left-temporal Words ≈ Non-words ($|dz| < .5$, TOST $p = .03$, BF01 = 4.2). **Evoked** 250–600 ms, the reverse: left-temporal W > NW ($dz = 1.04$, $p < .001$), occipital n.s.

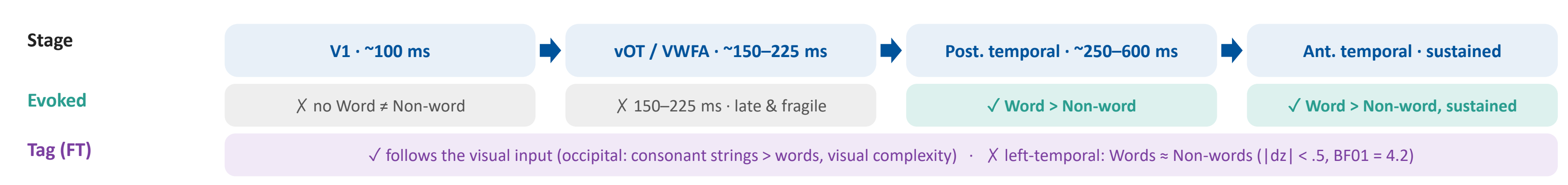
N = 20 (sensor cohort), same epochs for both measures; per-word SNR, 2.83 s FTT, 3 neighbours/side; diamonds = mean ± 95 % CI; Tag × Wordness n.s.

4 · Direct test: the word-locked response beats the tag, ROI by ROI (Aim 2)



Direct test: the **word-locked response** beats the **tag** on left-temporal sensors ($Adz = 1.02$, $p = .01$); the tag stays near zero along the pathway (all n.s.).

N = 17 (source cohort), same per-word epochs; bootstrap 95 % CI; filled = $p_{\text{FDR}} < .05$. 80–130 / 150–225 ms windows: QR.



Aim 1 — partly: temporal stages replicated, vOT only late and fragile, latency order n.s. **Aim 2 — no:** the tag indexes visual input, not wordness; the word-locked response does.

References

- Salmelin, R. 2007. "Clinical Neurophysiology of Language: The MEG Approach." Clinical Neurophysiology 118, no. 2: 237–254. <https://doi.org/10.1016/j.clinph.2006.07.316>.
- Samoylov, A., Semenova, N., Myachikov, A., & Shtyrov, Y. (2025). Patterns of cortical activity in a silent single-word reading task depend on word frequency and age-related differences: An MEG study. Psychophysiology, 62(3), e70192. <https://doi.org/10.1111/psyp.70192>.
- Thesen, T., McDonald, C. R., Carlson, C., Doyle, W., Cash, S., Sherley, J., Felsovalyi, O., Girard, H., Barr, W., Devinsky, O., Kuzniecky, R., & Halgren, E. (2012). Sequential then interactive processing of letters and words in the left fusiform gyrus. Nature Communications, 3, 1284. <https://doi.org/10.1038/ncomms2220>.
- Cornelissen, P. L., Kringelbach, M. L., Ellis, A. W., Whitney, C., Holliday, I. E., & Hansen, P. C. (2020). Activation of the left inferior frontal gyrus in the first 200 ms of reading: Evidence from magnetoencephalography (MEG). PLoS ONE, 15(4), e023559. <https://doi.org/10.1371/journal.pone.0235559>.
- Norcia, A. M., Appelbaum, L. G., Ales, J. M., Cottareau, B. R., & Rossion, B. (2015). The steady-state visual evoked potential in vision research: A review. Journal of Vision, 15(6), 4. <https://doi.org/10.1167/15.6.4>.
- Figueira, J. S. B., Kutlu, E., Scott, L. S., & Keil, A. (2022). The FreqTag toolbox: A principled approach to analyzing electrophysiological time series in frequency tagging paradigms. Developmental Cognitive Neuroscience, 54, 101066.
- Kabdebon, C., Fló, A., De Heering, A., & Aslin, R. (2022). The power of rhythms: how steady-state evoked responses reveal early neurocognitive development. NeuroImage, 254, 119150.
- Lochy, A., Van Belle, G., & Rossion, B. (2015). A robust index of lexical representation in the left occipito-temporal cortex as evidenced by EEG responses to fast periodic visual stimulation. Neuropsychologia, 66, 18–31. <https://doi.org/10.1016/j.neuropsychologia.2014.11.007>.
- Wang, F., Kaneshiro, B., Strauber, C. B., Hasak, L., Nguyen, Q. T. H., Yakovleva, A., & McCandless, B. D. (2021). Distinct neural sources underlying visual word form processing as revealed by steady state visual evoked potentials (SSVEP). Scientific reports, 11(1), 18229.